

SOOHYUN CHOI

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning
Hanyang University · Information and Intelligence Systems Lab (IISL) · Seoul, Republic of Korea
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RESEARCH INTERESTS

Reinforcement learning and sequential decision making, with emphasis on multi-step policy improvement, analysis of tau regimes, offline goal-conditioned RL, and generative long-horizon state-space planning.

EDUCATION

Hanyang University <i>Integrated M.S./Ph.D. Program in Electronic Engineering</i> Information and Intelligence Systems Lab (IISL) · Advisor: Prof. Songnam Hong	2024-Present
Hanyang University <i>B.S. in Electronic Engineering</i>	2018-2024

MANUSCRIPT & SELECTED RESEARCH

- S. Choi, S. Cho, and S. Hong**, "PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning." *Manuscript, 2026*.
- Developed explicit state-space bridges, inverse-dynamics action chunks, and execution-time replanning for long-horizon offline GCRL; evaluated on OGBench navigation and manipulation tasks.
- S. Choi, M. Oh, and I. Song**, "ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian Manifold." *Undergraduate Capstone Technical Report, 2024; revised 2026*. [\[PDF\]](#)
- Formulated designer-selected performance equivalence classes for circuit search and combined the induced quotient geometry with attention-weighted line search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

Multi-step Proximal Policy Improvement 2026-Present

- Analyze how proximal step size tau governs recursively re-centered policy refinement, using sweeps and mathematical displacement diagnostics to characterize local-flow, nonlocal finite-step, and over-refinement regimes.

PathFlower - Goal-Conditioned State Flows Current

- Extend PathBridger from an explicit subgoal and bridge toward final-goal-conditioned state flows that generate useful intermediate states for long-horizon planning.

SK hynix-Hanyang University Industry-Academic Research Project 2024-2025

- Contributed to machine-learning- and transfer-function-based modeling for circuit-interconnect waveform and delay estimation, including model validation and industry technical exchanges.

HONORS & FUNDING

- Outstanding Graduation Project Award - KRW 2,000,000 award
- Excellence Award, College of Engineering Capstone Design - KRW 1,000,000 award
- BK Research Assistantship (BK-RA), Hanyang University - 2025

SELECTED GRADUATE COURSEWORK

Learning & Control: Machine Learning; Robot Learning; Advanced Optimal Control Theory
Probability & Analysis: Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I